

PRITISH SAHA

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Education

Indian Institute of Technology, Kharagpur

Kharagpur, West Bengal

B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering

Expected completion: May'27

Relevant Courses: Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra

Research & Engineering Experience

Research Fellow (MARS 4.0) | Cambridge AI Safety Hub | UK (Hybrid) | [\[Report\]](#) Dec'25 – Present

With Prof. Fernando Rosas, studying Bayesian belief representations in transformers and extending the experiments to RL

- Constructed **exact Bayesian filters** for input–output transducers to study geometry under full and partial observation
- Decoded fully observed Bayesian beliefs at $R^2 = 0.985\text{--}0.997$ in four-layer transformer pilots trained for 100k updates
- Added exact **KL-regularized POMDP** solvers and reward-switch assays, with **115 tests** of geometry and control

Data Science Intern | Axis Bank, Business Intelligence Unit | Mumbai | [\[Report\]](#) May'26 – Jul'26

Built a PySpark graph analytics pipeline over 224.8M accounts and 1.31B transfer edges to identify personal-loan fraud rings

- Pruned the working graph $\sim 20\times$ using time-respecting 3-hop BFS, retaining connectivity for **$\sim 80\%$ of applicants**
- Implemented **weighted PageRank** and **Bahmani peeling** in PySpark; used HDFS checkpoints to recover graph jobs
- Delivered an explainable four-signal review queue, with **5.49 $\times$ mean lift** in the **top 1%** across three monthly cohorts

Research Intern | RAAPID INC | Remote | [\[Research\]](#) [\[Patent\]](#) [\[LoR\]](#) Mar'25 – Jul'26

With Prof. Amitava Das, developed GRIT for efficient LLM adaptation and built quality checks for clinical entity extraction

- Prototyped **Triton kernels** for rank-space covariance, damped Cholesky inversion and LoRA gradient preconditioning
- Integrated **CUDA streams/events** and PyTorch fallbacks for async K-FAC inversion and Fisher eigendecomposition
- Applied NER rules to stored predictions: **0.640 exact micro-F1** and **zero subword fragments** across 5,035 test rows

Research Intern | Complex Networks Research Group, IIT Kharagpur | [\[Report\]](#) [\[LoR\]](#) Jul'25 – May'26

With Prof. Pawan Goyal, built LaViDA to investigate latent reward shaping for mathematical reasoning in Qwen2.5-Math-7B

- Built **single-H100 GRPO** with LoRA, TRL and vLLM, using **16 rollouts per prompt** and frozen-VAE features
- Evaluated a selected Oracle-augmented MSE pilot: **79.57% MATH-500** mean correctness ($n = 8$; **+4.70 pp** vs GRPO)
- Found shaped and binary advantages nearly aligned (cosine **0.979**), with no shaping signal on all-wrong rollout groups

Publications & Preprints

Mixture of Chapters | ICLR 2026 NFAM Workshop | [\[Code\]](#) | [\[Paper\]](#) Mar'26

Co-authored “Mixture of Chapters: Scaling Learnt Memory in Transformers”, accepted at the ICLR 2026 NFAM Workshop

- Introduced **cross-attention** over **262K memory tokens**; sequences use 64 routed chapters and one shared chapter
- Pretrained **150M–1B-parameter models** on up to **15B tokens**, testing memory-augmented and dense transformers
- Observed ARC-C forgetting of **2.68 pp** after instruction tuning, versus **6.69 pp** for the iso-FLOP transformer baseline

GRIT: Geometry-aware PEFT | arXiv Preprint | [\[Code\]](#) | [\[Paper\]](#) Jan'26

First-author preprint on second-order LoRA adaptation, combining curvature-aware updates with dynamic rank allocation

- Combined **rank-space K-FAC** preconditioning and Fisher-guided basis updates to adapt LoRA ranks in fine-tuning
- Reached **66.19% on GSM8K** vs LoRA's 63.38% with Llama-3.1-8B and **27.8% fewer** effective adapter parameters
- Exported a **32% smaller adapter** with **<0.04 pp** accuracy loss; evaluated five tasks on 3B and 8B Llama backbones

Competitions

3rd Place | General Championship Data Analytics, IIT Kharagpur | [\[Report\]](#) Mar'26

Captained Patel Hall's team, developing Frammer AI's analytics dashboard with text-to-SQL and automated KPI discovery

- Developed SQL generation with retrieval and repair: **90% execution accuracy** on a **50-query BIRD** mini-dev sample
- Automated KPI discovery with a **nine-stage LangGraph** pipeline; tested on **874K synthetic rows** across 58 profiles

Overall Champions, IIT Kharagpur Contingent | Inter IIT Tech Meet 14.0 | [\[Report\]](#) Oct'25 – Dec'25

Co-designed DRISHTI for ISRO GeoNLI, a remote-sensing system for captioning, grounding and VQA across RGB/SAR/IR

- Designed a **Qwen3-VL-8B** curriculum with **LoRA SFT and DPO**, from general adaptation to captioning and VQA
- Paired **SAM3 pyramidal tiling** with VLM mask refinement for object counting, area estimation and spatial grounding

Participant | Amazon ML Challenge 2025 | [\[Code\]](#) Oct'25

Scored **40.8 SMAPE** on product pricing by stacking a fine-tuned Qwen2.5-VL-3B with LightGBM on image and text features

- Processed inputs offline and streamed tensors with **WebDataset**, raising **4-bit QLoRA** training throughput by **$\sim 75\%$**
- Fixed VLM special-token masking; fit LightGBM in price space with **Pseudo-Huber loss** and monotonic constraints

National Finalist | American Express Campus Challenge, Decision Science | [\[Report\]](#) Jul'25

Reached **0.59 MAP@7** on the final evaluation set by combining LightGBM, XGBoost and a transformer for offer ranking

- Generated **3K+ customer-history features** in parallel with point-in-time joins and customer-grouped cross-validation
- Trained a transformer on **GBDT residuals** with a **listwise ranking loss** to correct errors left by the tree ensemble

Technical Skills

- Languages & Tools:** Python, C++, SQL, CUDA, PySpark, Hadoop, Linux, Docker, SLURM, Git, vLLM, LangGraph
- ML Stack:** PyTorch, JAX, Triton, FSDP/DTensor, Transformers, FlashAttention-2, PEFT/LoRA, bitsandbytes, TRL

Extra-Curricular Activities

Karate black belt and tournament participant | Patel Hall interhall football and water polo teams | NSS village volunteer